

“A Review of Magnetic-Field Sensing Technologies: Applications in Energy, Transportation & Industry”

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Abstract (194 words):

Magnetic-field sensors enable non-invasive measurement of electrical current and magnetic fields in energy, transportation, and industrial systems, including grid transformers, automotive drivetrains, and robotic motion-control systems. These measurements support motion monitoring, material characterization, and equipment condition assessment for prediction, control, and forecasting. Their use can improve reliability and efficiency while reducing costs. Understanding sensor operating principles, transduction mechanisms, and operational limits is essential for identifying suitable applications and assessing the technical value of each technology. This review compares technologies from established Hall-effect sensors to emerging quantum magnetometers based on nitrogen-vacancy centers in diamond. It examines their integration with artificial intelligence (AI) tools and relates performance and commercial maturity to representative applications in these domains. The review also evaluates data-driven magnetic sensing for real-time and in situ monitoring, digital twins, and adaptive decision-making under dynamic conditions. An integrated sensing-system approach can exploit the strengths of each technology to improve measurement quality and system capability. Future progress requires closer integration of sensing hardware and computation while maintaining traceability and compliance with relevant standards and regulations. By connecting sensing principles with application value and technology readiness, the review provides a practical framework for magnetic-sensing development and research.